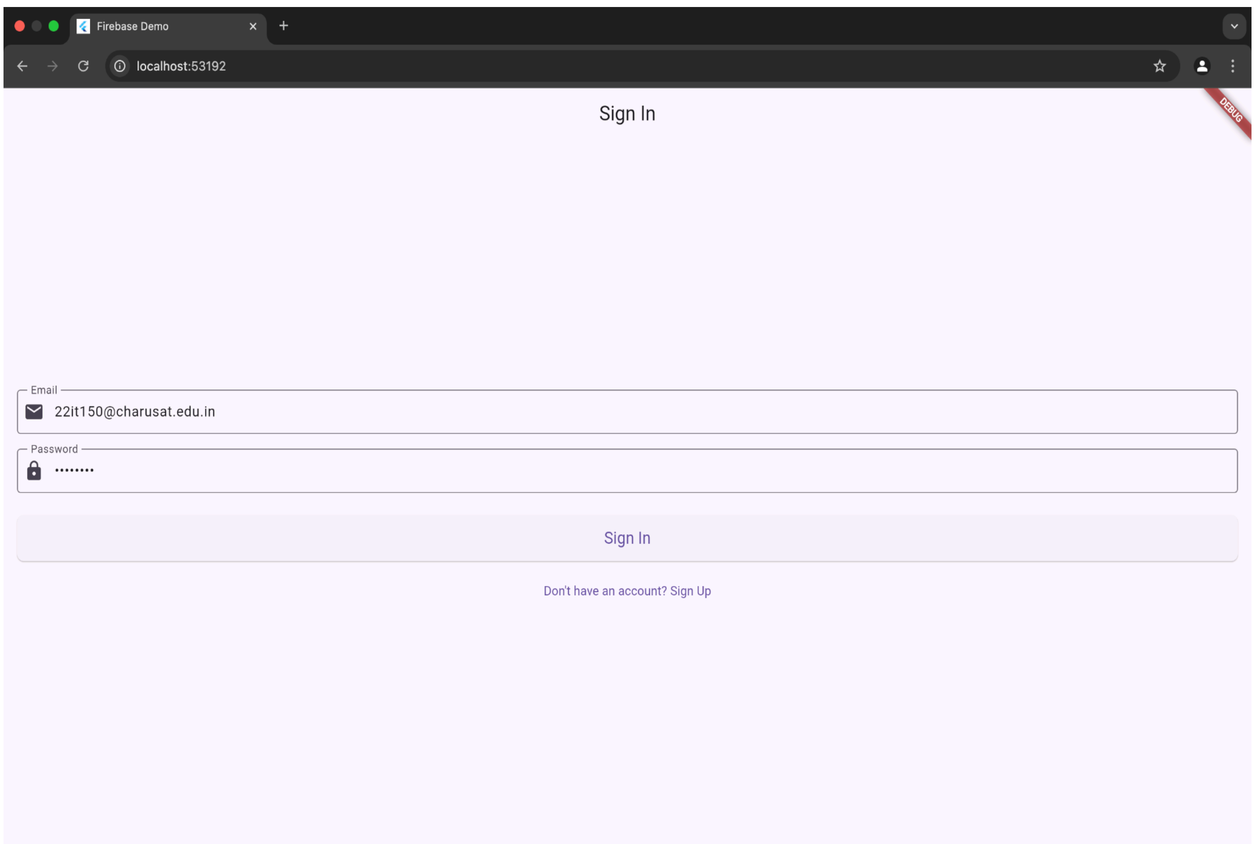
Problem Statement
At SmartFab Industries, our production teams currently rely on paper logs and ad-hoc spreadsheets to record raw material usage, compute manufacturing costs, and set selling prices,
leading to stock discrepancies, cost-overruns, and slow, error-prone reporting. To streamline operations and boost profitability, we need a Flutter-based Material Tracking & Costing App that enforces role-based access, captures real-time consumption via QR/barcode scanning, automates all cost calculations, and keeps inventory and cost histories in sync across devices and the cloud. The app must let Admins manage materials, processes, users, and view analytics, while Operators can only scan materials, log usage, and see assigned tasks. It should work offline—caching scans and logs locally—and reconcile seamlessly with Firebase (or similar) once online. Your task is to deliver a working prototype that addresses these challenges while following Flutter best practices for state management, offline persistence, and secure data handling.
Key Pain Points to Solve:
Manual material logging causing stock mismatches and lost data.
Time-consuming, error-prone cost and price calculations.
Lack of role segregation leading to unauthorized edits or data leaks.
Delayed visibility into low-stock alerts and cost overruns.
The ideal solution will balance performance, usability, and security while demonstrating clean, well-structured, and maintainable code.
Note:
Role-Based Access Control (RBAC):
o Admin: Add/Edit/Delete materials, processes, users; view dashboards, analytics, and exportable reports.
o Operator: Scan materials, log consumption, and view only assigned operations.
Real-Time Material Tracking:
o Scan QR/barcodes to auto-fetch material details (name, unit cost, unit type, stock).
o Offline caching of scans/logs in Hive or SQLite; sync changes on reconnection.
Automated Cost Calculations:
o Raw Material Cost = Unit Cost × Quantity Used
o Manufacturing Cost = Raw Material Cost + Additional Processing Costs (labor, energy, etc.)
o Final Product Price = Manufacturing Cost + Desired Margin
o Suggested Selling Price and Profit Margin computed per unit.
Inventory Management:
o Real-time stock level updates on each consumption log.
o Low-stock alert notifications for Admins.
o Maintains a searchable history of consumption by date, material, and product.
Reporting & Exports:
o Cost breakdown reports (raw material, processing, selling) viewable by Admin.

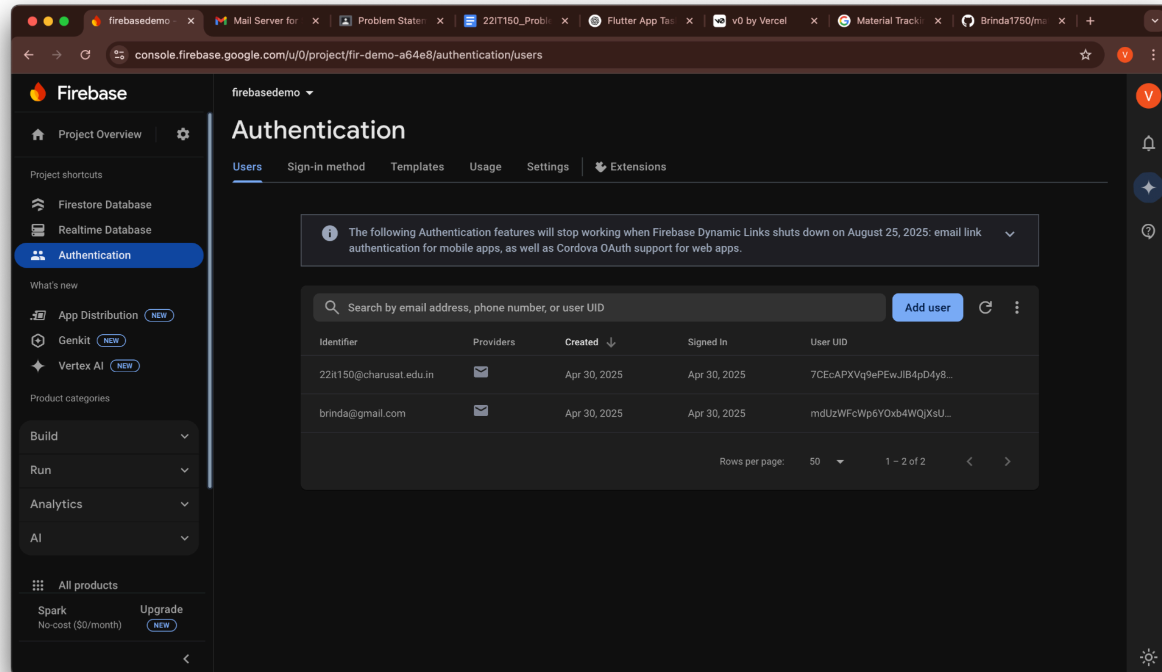
IT366 Mobile Application Development
Practical Question Paper

Semester: 6th Semester B. Tech. (IT)
Date: 30/04/2025 (Wednesday) Time: 01:00PM TO 04:00PM

o Export logs and summaries as PDF or CSV for further analysis.
Data Storage & Sync:
o Local persistence (Hive/SQLite) for instant access and offline use.
o Cloud sync (e.g., Firebase Firestore) for backup, multi-device consistency, and real-time updates.

SCREENSHOT:




console.firebase.google.com/u/0/project/fir-demo-a64e8/authentication/users
Firebase
Project Overview
Project shortcuts
Firestore Database
Realtime Database
Authentication
What's new
App Distribution NEW
Genkit NEW
Vertex AI NEW
Product categories
Build
Run
Analytics
AI
All products
Spark
No-cost ($0/month)
Upgrade NEW
firebasedemo
Authentication
Users
Sign-in method
Templates
Usage
Settings
Extensions
The following Authentication features will stop working when Firebase Dynamic Links shuts down on August 25, 2025: email link authentication for mobile apps, as well as Cordova OAuth support for web apps.
Search by email address, phone number, or user UID
Add user
Identifier
Providers
Created
Signed In
User UID
22it150@charusat.edu.in
Apr 30, 2025
Apr 30, 2025
7CEcAPXVq9ePEwJlB4pD4y8...
brinda@gmail.com
Apr 30, 2025
Apr 30, 2025
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Rows per page: 50
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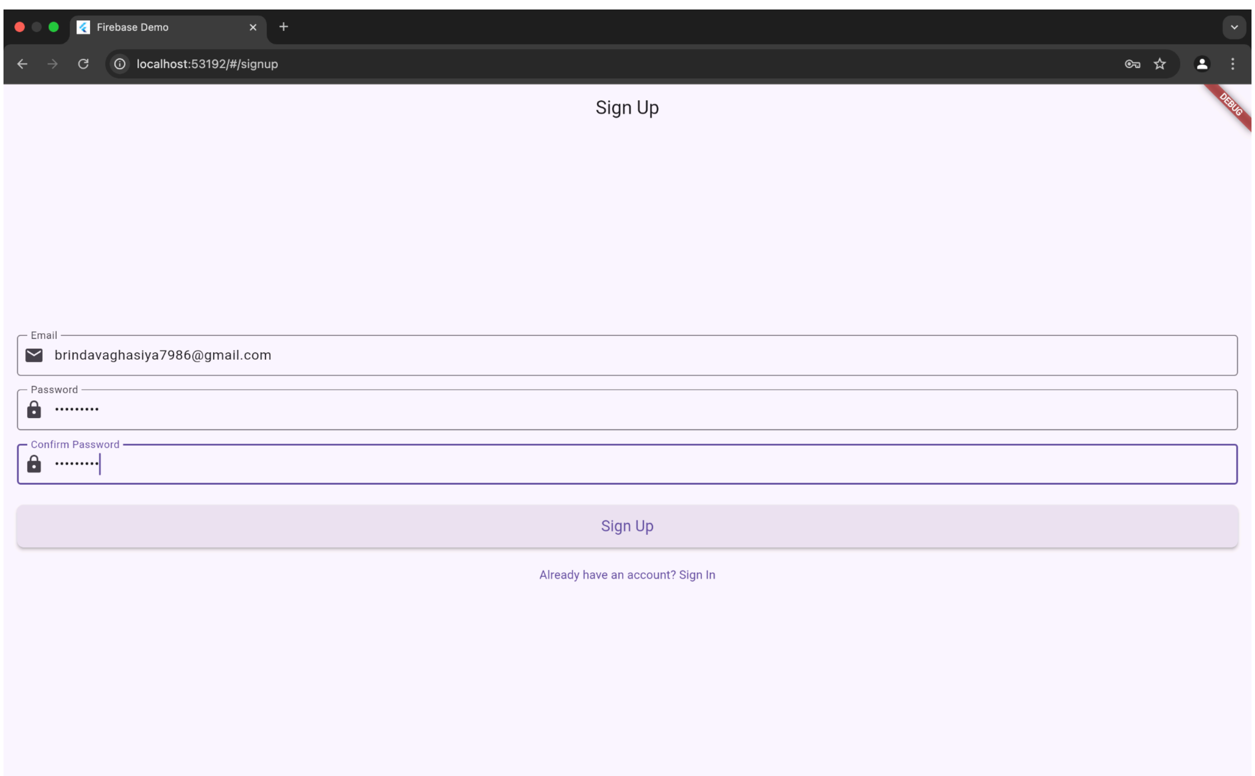
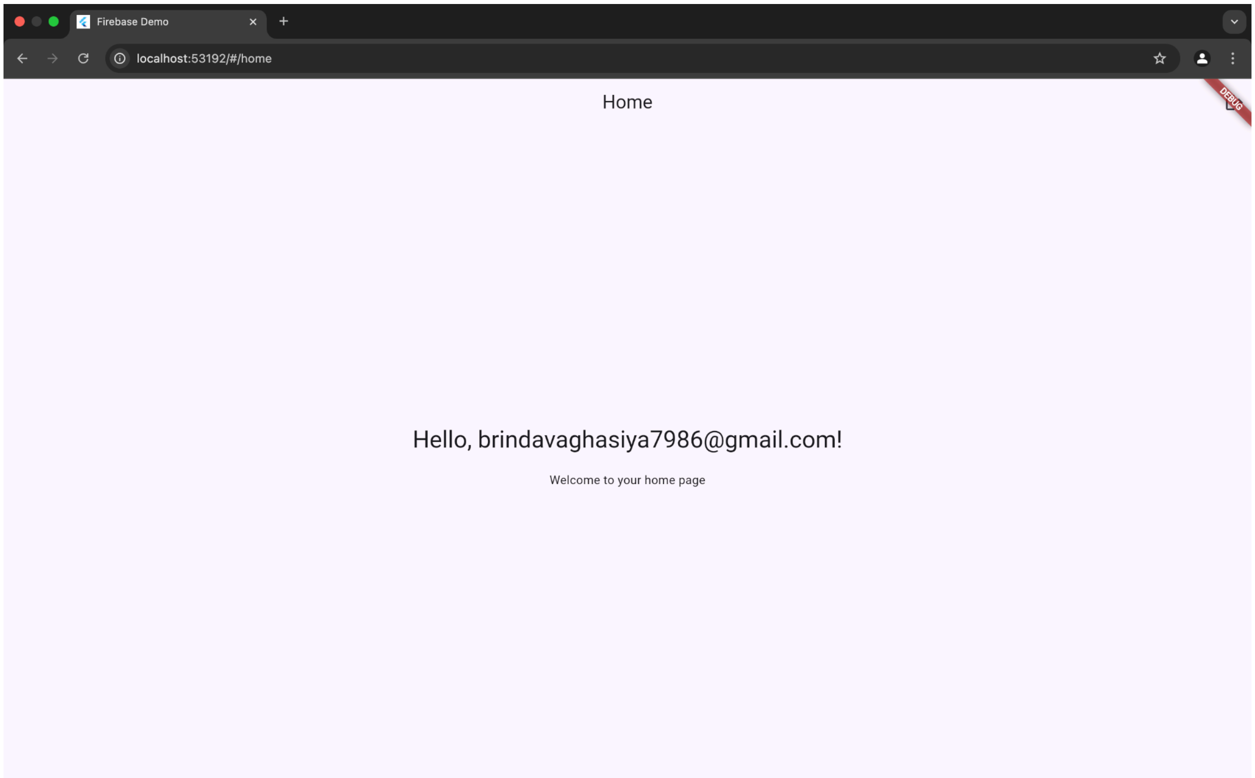
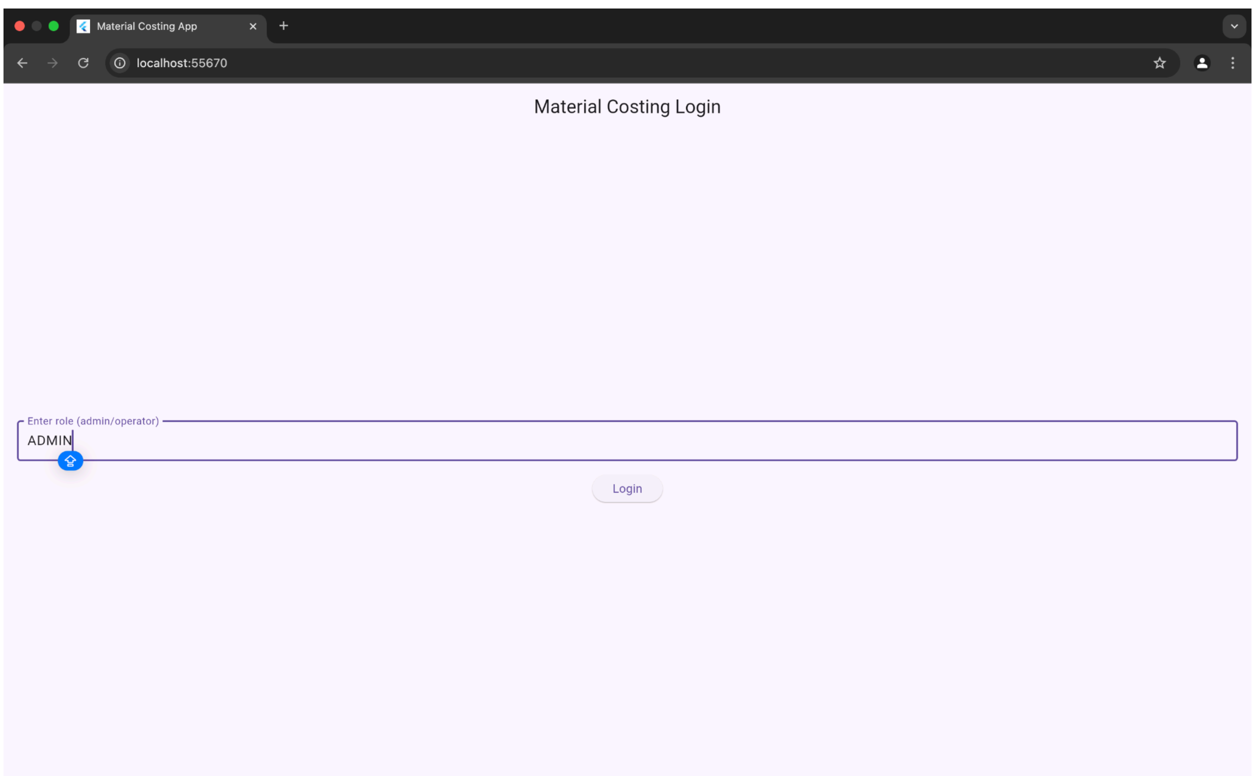
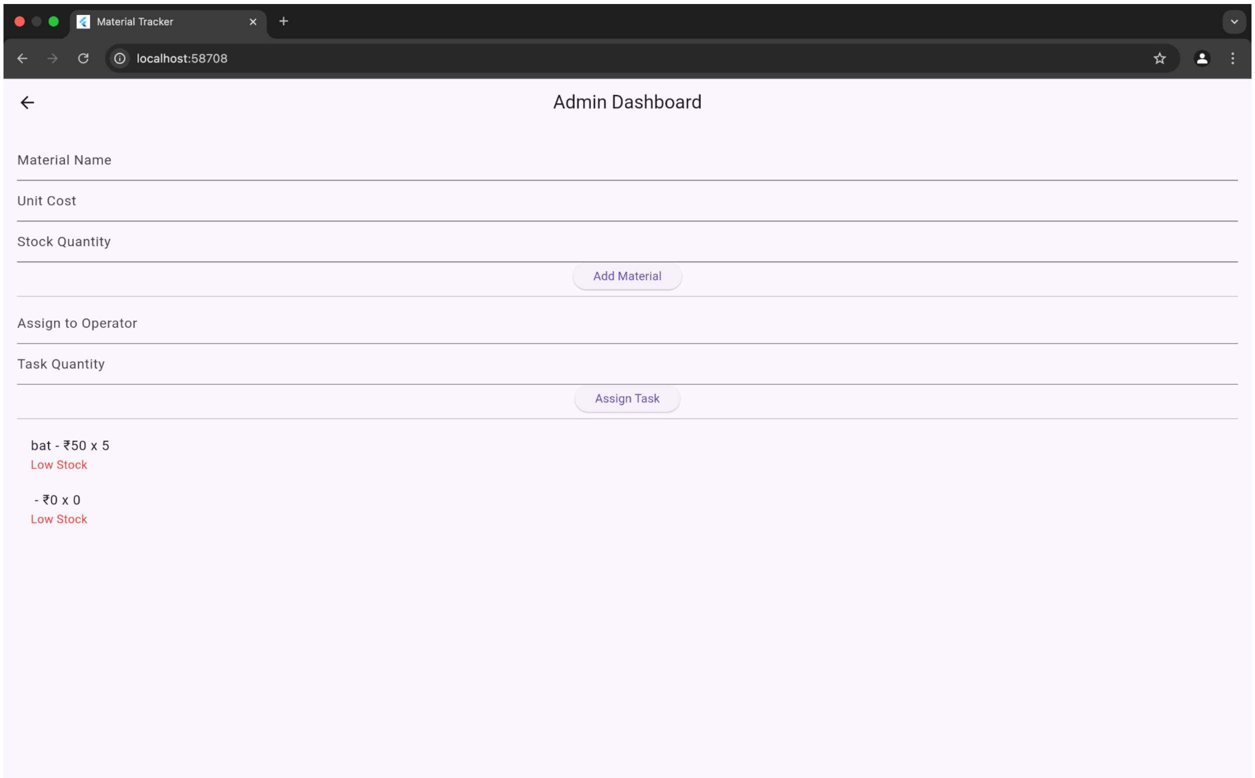
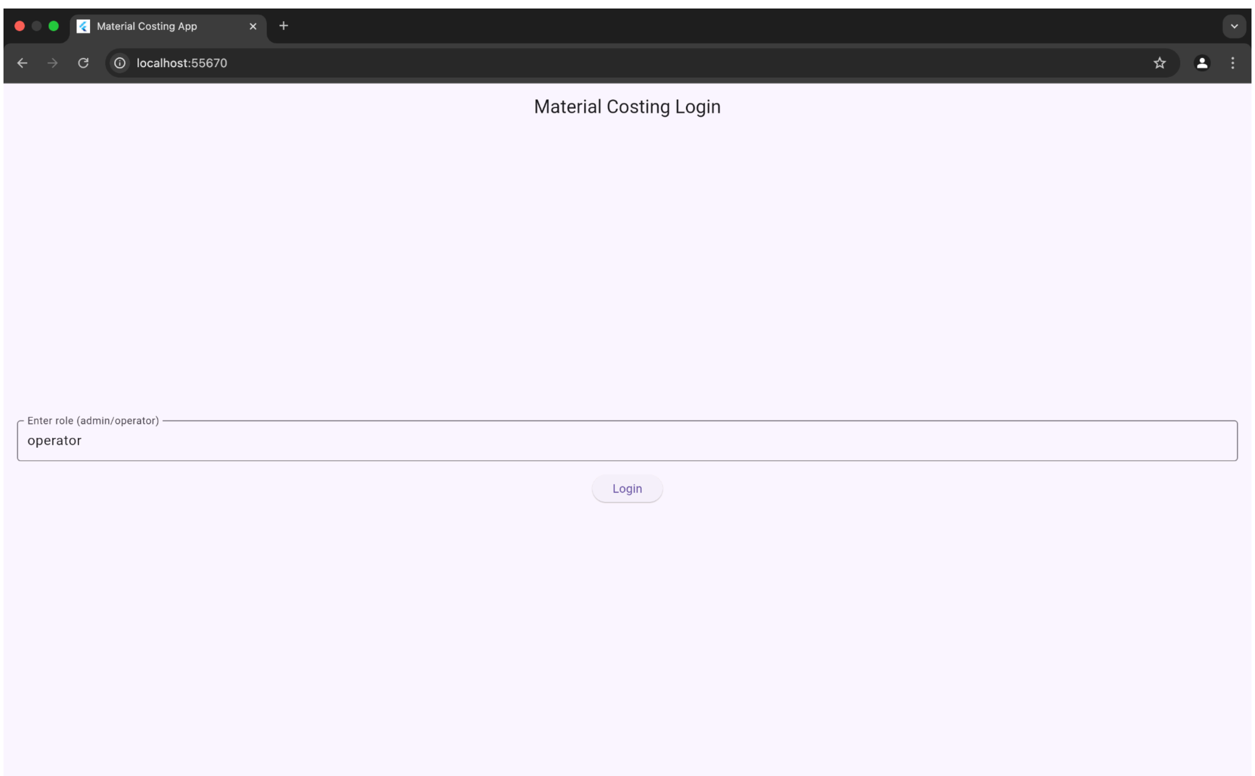
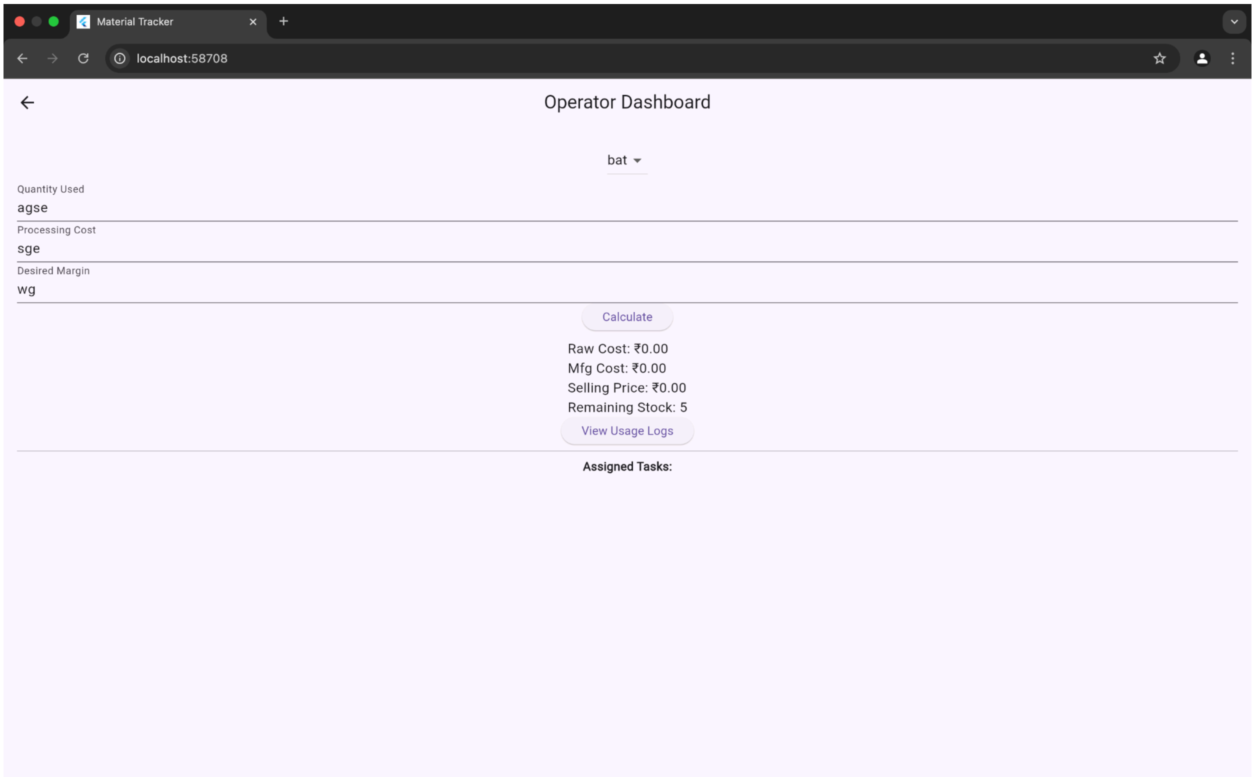

Firebase Demo
localhost:53192/#/signup
Sign Up
DEBUG
Email
brindavaghasiya7986@gmail.com
Password
•••••••••
Confirm Password
•••••••••
Sign Up
Already have an account? Sign In

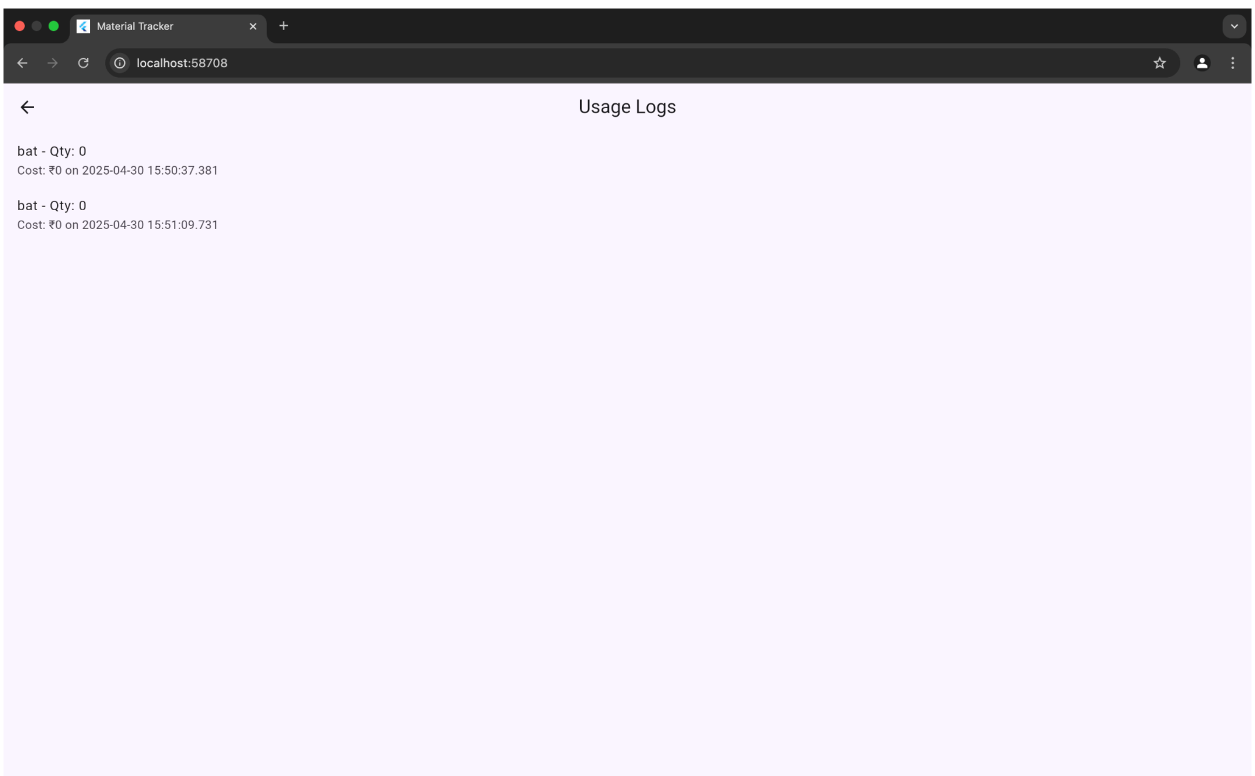

Firebase Demo
localhost:53192/#/home
Home
DEBUG
Hello, brindavaghasiya7986@gmail.com!
Welcome to your home page

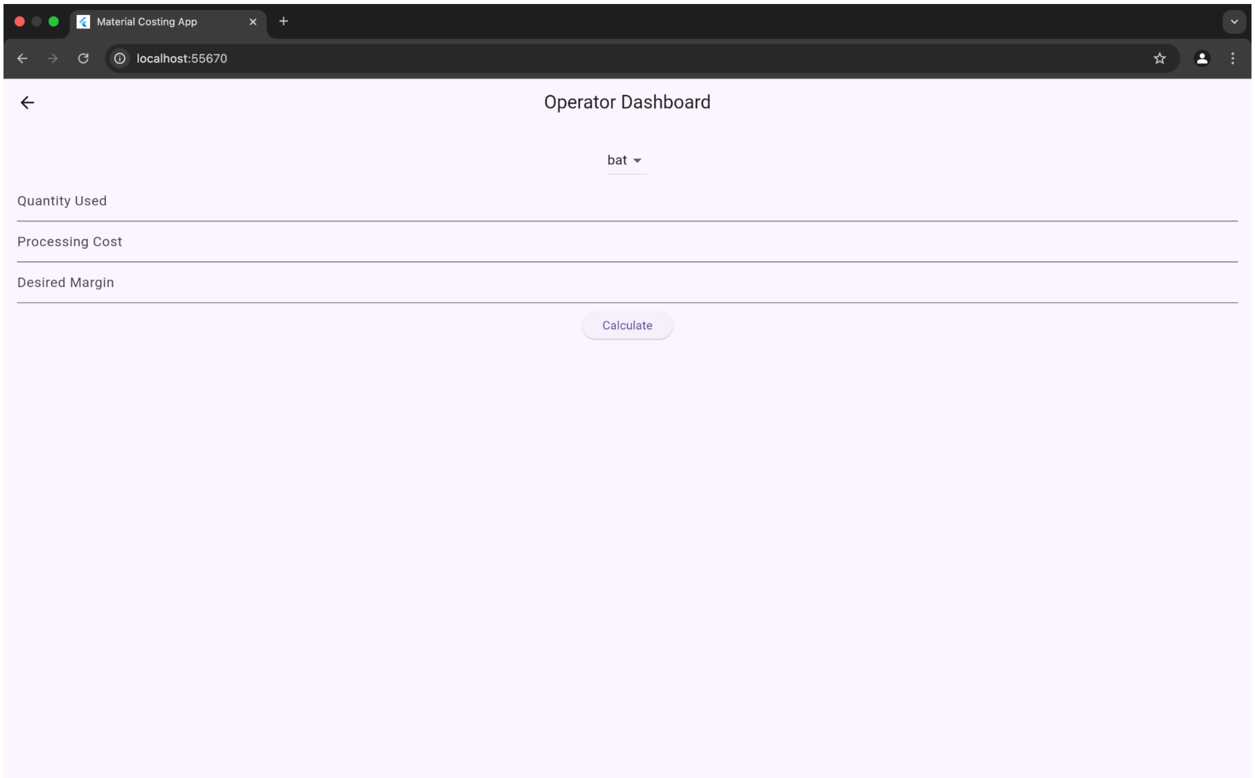

Material Costing App
localhost:55670
Material Costing Login
Enter role (admin/operator)
ADMIN
Login

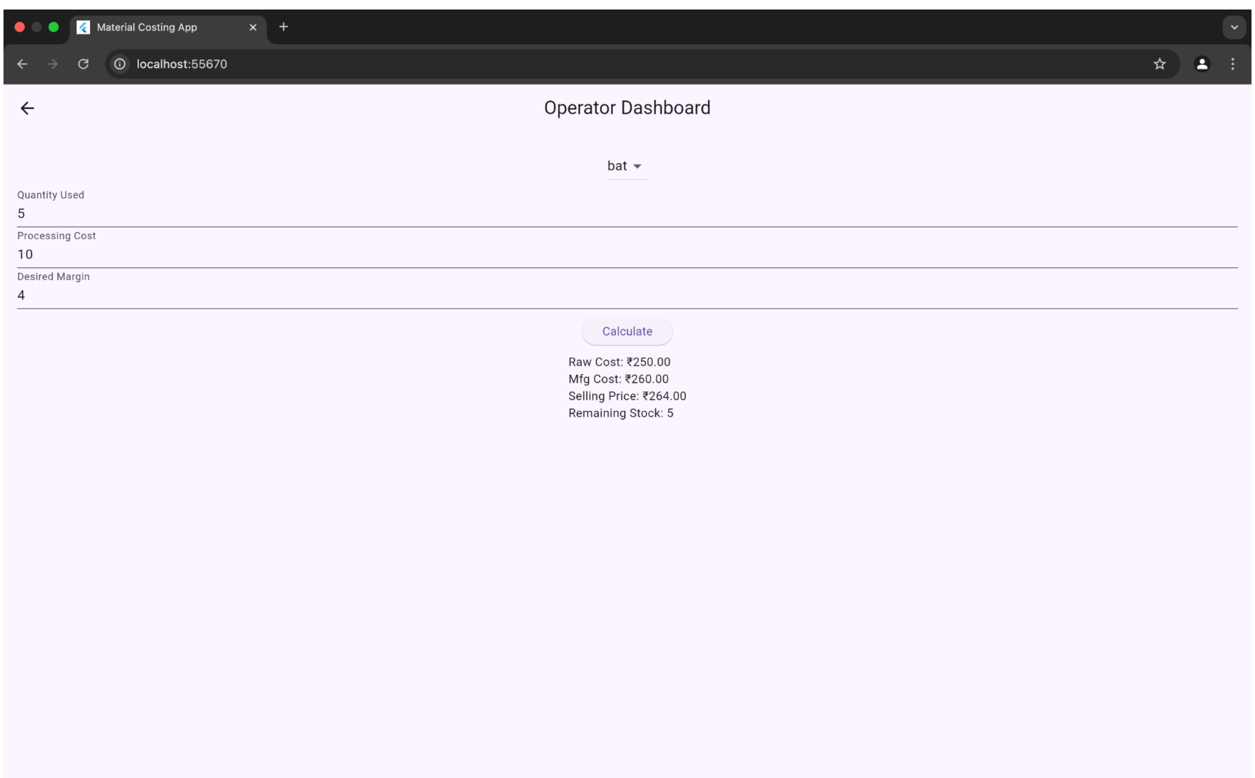

Material Tracker
localhost:58708
Admin Dashboard
Material Name
Unit Cost
Stock Quantity
Add Material
Assign to Operator
Task Quantity
Assign Task
bat - ₹50 x 5
Low Stock
- ₹0 x 0
Low Stock


Material Costing App
localhost:55670
Material Costing Login
Enter role (admin/operator)
operator
Login


Material Tracker
localhost:58708
Operator Dashboard
bat
Quantity Used
agse
Processing Cost
sge
Desired Margin
wg
Calculate
Raw Cost: ₹0.00
Mfg Cost: ₹0.00
Selling Price: ₹0.00
Remaining Stock: 5
View Usage Logs
Assigned Tasks:


Material Tracker
localhost:58708
Usage Logs
bat - Qty: 0
Cost: ₹0 on 2025-04-30 15:50:37.381
bat - Qty: 0
Cost: ₹0 on 2025-04-30 15:51:09.731


Material Costing App
localhost:55670
Operator Dashboard
bat
Quantity Used
Processing Cost
Desired Margin
Calculate


Material Costing App
localhost:55670
Operator Dashboard
bat
Quantity Used
5
Processing Cost
10
Desired Margin
4
Calculate
Raw Cost: ₹250.00
Mfg Cost: ₹260.00
Selling Price: ₹264.00
Remaining Stock: 5

Future Work

- **Add offline storage using Hive/SQLite.**
- **Integrate Firebase for real-time sync and cloud backup.**
- **Implement QR/barcode scanning for material tracking.**
- **Add secure login and role-based access control.**
- **Enable PDF/CSV export of reports.**
- **Show analytics like cost trends and low-stock alerts.**
- **Assign tasks to Operators for better workflow.**

Conclusion

This app provides a simple solution for tracking materials and calculating manufacturing costs using Flutter. It supports role-based access for Admin and Operator, allowing secure and efficient operations. With further improvements like cloud sync and QR scanning, it can evolve into a complete industrial solution.

https://github.com/Brinda1750/material_tracker/tree/main